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AI-Based Legal Evidence Authenticity Scoring and Forgery Detection

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Abstract*— The use of scanned and digitally stored documents in legal, government and administrative systems has raised the possibility of tampering and forging documents. The traditional manual verification approaches can be slow, subjective and fail to identify any manipulations at the pixel level. In order to overcome these shortcomings, this paper proposes an automated system of document authenticity verification, combining deep learning algorithms with image forensic verification. The proposed system uses Error Level Analysis (ELA) to detect the compression inconsistencies that can be used as evidence of tampering, and it is refined by a convolutional neural network (CNN) to detect the exact location of the forged areas at a fine-grained level. Besides, metadata consistency validation like document creation date, history of modifications, and information about devices are also conducted in order to increase the forensic reliability. The major input to the system will be a visual forgery heatmap that shows suspicious areas, and an objective Authenticity Score of 0 to 100 that measures the chances of document integrity. Integrating visual interpretability with numerical evaluation, the suggested method will give qualitative and quantitative data, which will support scientifically valid and legally admissible decision-making. The system is also expected to minimize human error, enhance the efficiency of verification as well as a reliable digital document validation system to be used in real life legal and administrative application.

Keywords—Image Forensics, Deep Learning, Legal Document Verification, Convolutional Neural Network, Error Level Analysis, Document Security, Authenticity Scoring.

I. INTRODUCTION

A. The surging threat of Document Forgery.

The legal documents are also contractual, ID records, certificates and financial instruments, all of which would form the foundation of the administrative mechanism and the judicial system of the majority of countries worldwide. The documents have been facilitated by the workflow globalization making the digitization of the documents even faster process to enable faster storage, transmission and processing of the documents. This has however changed with other areas of weakness being generated. A high level of accuracy can now be achieved in manipulating revenue images through advanced image editing software that produces forgeries, which can not to be defined as the original of the forgery[1][4][12].

These forgeries are either in the format of a simple manipulation of the text up to sophisticated copy-move forgeries and splicing[5][16][19]. This is a menace to the national security, money and the law[11][12]. Even the smallest of things like the rescheduling of a date, signature or a clause omission can be mammoth proportions in the court-room.

B. *Weaknesses of existing Checking Systems.*

The current processes of document validation can be clustered into two and both protocols possess a drastic demerit .The most pervasive is the hand held one. Papers are compared by human analyser's with reference model papers and in a bid to find visual differences[1][10]. It is not dynamic, professional and cannot be extended to this degree as Know Your Customer processing (KYC) or e-governance portal is. The most significant is that the manipulations of pixels (noise injection, compression artifacts, and localized retouching) that have been implemented by the modern editing programs and are undetected leave no trace whatsoever can never be tracked down by human experts in the domain[3][4][9].

Rule based checks, checksum checks and simple metadata checks are the format checks that are common to the typical automated testing tools[2][10]. These methods are vulnerable to adversarial and invisible attacks of manipulation and can identify signature based forgeries only. Neither will either manual, or the least automated system provide:

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Localization: identifying what was done in the documents.

Quantification: the creation of gauging of certainty of admissibility of law.

Interpretability: which gives visual evidence, which can be understood by non-experts.

The non-existence of such functions renders the weight of evidence of the digital documents in a court case meaningless and spoils the administrative systems in a systematic manner[8][12][14].

C. Proposed Approach and Contributions.

In the next presentation we offer an automated forensic system that fills out these lapses through the combination of three aspects:

- The different compression discrepancies are detected by means of Error Level Analysis (ELA) and the differences are viewed as the sign of the tampering with the data[1][4].
- The convolutional neural network that is based on U-Net detects[7][8][15] pixels that are forged .Metadata provenance Checking provides document provenance and modifies artifacts[10][21].
- Two similar outputs will be generated by the system: Data of visual forgery heatmap[8][9][14]. A measure of 0-100 of Authenticity of documents in terms of integrity.
- It is a two outputs test design that does not only have the capability of being investigated by the human eye to be able to do so but also has a quantitative evidence that meets the requirement of the legal admissibility condition[11][12].

The following are the recommendations:

1. Both ELA based feature enhancement and CNN based segmentation combined as forensic pipeline [1][4][6].
2. A variated U-Net architecture to the localization of document forgery [7][15].
3. Multi-dimensional scoring system of the pixel-based forensics framework and metadata analysis [12][21].
4. Extensive tests on 15, 000 document images have been carried out in many forms of forgery [8][17][18].
5. The training/validation protocols would be published publicly to allow them to be reproducible [16][19].

D. Paper Organization

Section II entails a literature review, which is linked to image forensics and deep learning-based detection. Section III will be a description of our methodology which will include preprocessing, network architecture and scoring. The fourth section will be the experimental set up, results and discussion. In section V, the mandates and instructions are filled in.

II. LITERATURE SURVEY

A. Passive Image Forensics

The passive image forensics is the study of the natural marks created in the process of manipulation which do not imply the presence of any embedded watermark or the digital signature[1][4]. The idea that is constructed out of these techniques is that real pictures have certain statistical features which are disturbed when they are perturbed[1][4].

These are some of the tools of the passive form of forensic analysis that are typically used, the Error Level Analysis (ELA)[1]. ELA exploits compression artefact of the JPEG compression and an image of a particular quality is saved and the disparities between the pixel values are made. On the aspects of the manipulated images, there exists the asymmetric distribution of errors as opposed to original areas in the case of manipulation happening in the course of the process[1][2]. The equation of ELA map E is calculated as: E =(8001-100)/100I)/100); mathematically it implies that an input image I must be provided.

$$E(x,y) = |I(x,y) - JPEG_Q(I)(x,y)|$$

JPEG compression reached by recompression with quality factor Q= JPEG Q-85 = (95-85) points to JPEG quality. It is also a good job that ELA has done because a damaged area would not possess an absolute copy of original compression properties. However, it cannot be considered as ELA sufficient[1][9][15]. It gives spurious posit on images with non-homogeneous histograms of compression and can be fooled by even advanced forgers who recreates compression artifacts. Besides that, ELA produces the qualitative heatmaps that are to be read by the experts, rather than providing the quantitative confidence, and automatically does the localization. The other forensic evidence is the aberration of the patterns of noises, deviation of camera response functions and demos icing effects [2,3,4]. It has the weakness of providing one with a hybrid solution with the individual efforts of the solutions.

B. Deep Learning to identify Forgery.

First forgery detectors were based on machine-based learning- based classical classifiers such as the Support Vector Machines [5][10]by hand. These techniques were not localised frequency-domain coefficients and statistical characteristics, e.g. noisy residues or edge artifacts .The CNNs had revolutionized the industry and innovated them. [6][7]

The methods of detecting forgeries in the current world are:

Classification: the identification of a picture of it is either real or not[6].

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Segmentation: small pixel masking .[7].The architectural designs through U-Net format have been useful in the localization of forgery in pixel-to-pixel basis. Another encoder-decoder architecture is U-Net, which has been developed based on the principle of the application of skip connections to retain the down sampling spatial data, which are lost. Some of the proposed variants were implemented in the forensic application like ManTraNet [9] and Constrained Neural Networks[10]. More precise, deep learning algorithms require huge quantities of data that are of heterogeneous nature. The inferences that have been extrapolated on the new forms of influence and reflect the results in a jurisprudential context are also challenging to make.

C. Document Verification System.

The most widespread commercial document verification processes are check of document by optical character recognition (OCR), checking of a barcode and cross-checking of a database[10][19].These strategies can only be used to compare the contents with the external materials but not such contents whose contents cannot be reproduced visually.

Document forensics research prototypes have been conducted by specialists. Digital analysis is done with physical properties of documents (paper fibre structure, ink analysis) in other combinations [11]. The provenance tracking can also be realized through the blockchain technology [12], which will never recognize forgeries, but only prevent the further forgeries.

D. *Gap Analysis and Positioning.*

It is acute, in omissions in the specified literature:

None of the systems that could be integrated presupposes the enhancement of functions with the help of ELA and being fragmented parts with the help of CNN. The standards of the practice are not present in the quantitative authenticity assurance of the legal admissibility .Document specific forensics is a field that has not received much research as compared to the natural image forging detecting .This is what we inject into it i.e. the proposal of hybrid ELA-CNN architecture on which we will impose the metadata addition and the quantitative scoring i.e. to certify legal documents.

Table I: Comparison of verification strategies.

System Category	**Strengths**	**Weakness**	**Output Type**
Manual Inspection	Human judgment, detects obvious fraud	Slow, error-prone, misses micro-forgeries	Qualitative (Expert opinion)
Basic Automated Tools	Fast, simple checks, low computation	Fails on advanced edits, no localization	Binary (Pass/Fail)
Traditional Forensics (ELA)	Finds compression anomalies, easy to read	False positives, defeated by re-saving	Feature Map (Qualitative)
Proposed Hybrid AI (ELA + CNN)	Scores + heatmap, fast,low error, multi-type	Needs large dataset, high training cost	Quantitative Score + Heatmap

III. METHODOLOGY

A.System Overview

These three stages pipeline are further referred to as the pipeline by fig 1. At the initial stage, the input documents are filtered with the assistance of ELA and their properties are enhanced. The resultant processed inputting forensic maps of features will be processed by a frame of a segmentation system created using U-Net that will lead to pixel wise hypothetical forgery mask. At the same time, the document provenance is also made with the help of metal data analysis module. Lastly is the component scoring that converts CNN output and CNN metadata anomalies and is expected to produce Authenticity Score, and visualization.

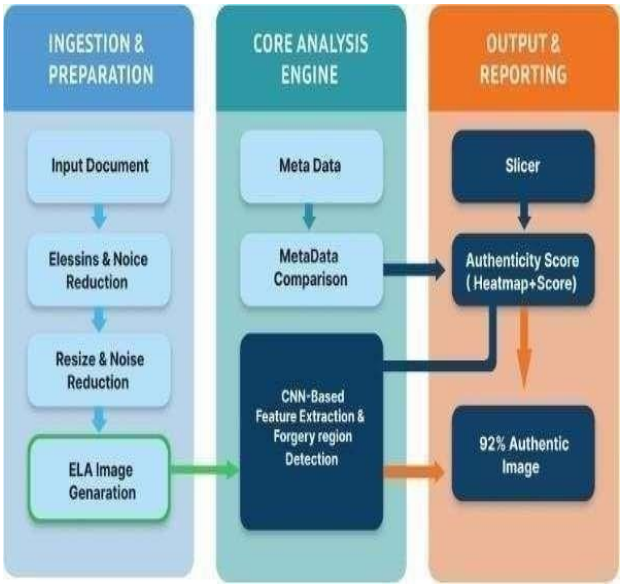


Fig.1 Document Forgery System Architecture

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B. Preprocessing and Additional Feature.

1)Image Normalization

It selects no things at all to whatever it may convey to the search to sift through the mass of the papers received :The rescaling of the 256x 256 size images is known as the rescaling Bilinear rescaling. It is an efficiency versus forensic description compromise .The input was fed by the RGB images and the three channel images respectively and the RGB images and the grayscale images respectively .Normalization of the intensities: It is brought to intensities [0,1].

2)Error Level Analysis

ELA transforms the forensic feature maps to 4 feature maps, provided by it in aid as contributing towards supporting the artifacts of the manipulation. We: in the image I normalised: we: S save/JPEG quality = Q = 95.And that of the withered looks I comp .Errors E = I – I comp. Scale E to [0,255] for visualization It has been chosen because of the finding, Quality=95, nearer the better the less compressible. The compression errors will be asymmetric and they will be appropriately considered in the CNN where the portions or areas that need to be manipulated will be rated high compared to the parts, which are not of interest .The current one being organized is the one, which is based on the Internet and is known as localization network. Network Forgery Nickel and Dimed.

a. Architecture: Modified U-Net

It has also been trained on U-Net optimum training as far as forensic segmentation is concerned fig 3. It is possible to determine the end of the architecture as shown in Table I below. Space dimensions will not go to waste because the padding will be used in all occasions to make up all the convolutional layers. The batches can also be normalized and hence, the training process presupposes the fastest training process besides the minimization of the overfitting. It is a dropout (rate=0.5) at each encoder block.

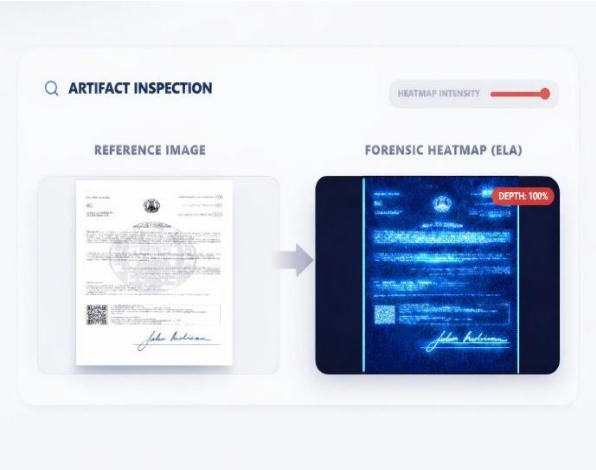


Fig. 2. ELA visualization example

b.Training Configuration

Loss Function: It is a loss in two components; A loss and a sum of Binary Cross-Entropy loss (BCE) and Dice coefficient:

$$L = L_BCE + l\, L_Dice$$

$$L_BCE = -\sum [y\, log(p) + (1-y)\, log(1-p)]$$

$$L_Dice = 1 - (2\sum yp + e) / (\sum y + \sum p + e)$$

We have selected l =1, e= 1/ 7, upon a particular numerical precision. This panacea is the equation of inharmony of the classes, (the regions, which are cut will have the greatest probability of containing less than 20 percent of document space), and the extent of space overlap which is the most desirable.

Optimizer: Learning rate=1e-4, β1=0.9, β2=0.999.

Batches size: 16 (maximum size due to memory graphics card)

Epoch: 50 and at early-stopping of learning (loss of validation= 10 on loss of validation)Data Augmentation To be sure that we are well generalizing we accept: Random rotation:+-15degHorizontal flip: 50 percent. Brightness adjustment: +- 10%Gaussian noise: σ = 0.01.

c. *Implementation*

With the help of the Tensorflow 2.13 / Keras, it is possible to apply this model. The training is done through the use of NVIDIA Tesla T4 and VRAM was 16GB which took approximately 8 hours to complete the 50 epochs

C. *Metadata Metanalysis.*

There is also metadata element filled with document properties which may be used to search the abuse:

The EXIF signatures are also lost, the trace of photo manipulations programs (adobe Photoshop, Gimp, etc.) are also lost. The signature of the scanned documents is not verified.

Timestamp Consistency: Timestamp documents may be checked on a time stamps. It is also in this that the flags get to know the impossibles; upon which is the day of rule of creation and the day of modification .The analysis is performed on internal object structure scanning to determine the abnormality of internal object structure including hidden layer, embedded object and broken cross-reference table retrieved by the analysis to form Pdf analysis and DOCX analysis .The above mentioned anomalies will be assigned the weight scores respectively in an effort to come up with the final score of the authenticity.

The second forecasting method is so-called approach of authenticity score:

A (Score Authenticity 0-100) is a combination of 3 items:

$$A = w1\ C_cnn + w2\ C_meta + w3\ C_ela$$

Where:

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- C_cnn = mean rate of average forgery between all pixels (inverted and scaled).
- C_meta = normalized metadata anomaly score (0 = lots of anomalies, 100 = beyond reproach)
- C_ela Confidence threshold analysis of ELA.
- w 1, w 2, w 3 are the weights that are experimentally determined (0.5, 0.3, and 0.2).

We scale to a ground truth set. The above 75 is regarded to be a genuine consideration and the points earned below 45 will be automatically deleted and medium score will be left to be considered by a human being.

IV. EXPERIMENTS AND RESULTS

A. Dataset Construction

We have developed Data Document Copy-move Dataset (DCD-15K) 15,000 document image-images.

TABLE II: Document dataset composition and forgery categories

Category	Count	Details
Authentic Documents	9,000	IDcards(2,500), contracts (2,500), receipts (2,000), certificates (2,000)
Capture method	-	60% smartphone, 40% scanner
Forged Documents	6,000	-
Copy-move	2,000	Textblocks duplicated within document
Splicing	2,000	Signatures/logos inserted from external sources
Retouching	2,000	Text removal or obfuscation

As per Table II Training: 10,500 (70%),Validation: 2,250 (15%),Test: 2,250 (15%).Stratification offers even handed forms of forgery of splits.

B. Evaluation metrics

They are specified to be founded on two levels of performances:

Localization accuracy: The accuracy of the localization pixel Intersection over Union (IoU),F1-score (harmonic average of the recall and the accuracy),Precision and Recall[14][17].

Image level measures are: Detection capability: Accuracy, Area Under ROC Curve (AUC),One to ninety-five false positive recall rate.

*C.*Quantitative Results

a). Main Results

Table III: Performance comparison.

Technique	***Pixel F1***	***IoU***	***Image Acc***	***AUC***
ELA thresholding	*0.412*	*0.261*	*0.703*	*0.712*
CFA-based	*0.567*	*0.398*	*0.811*	*0.843*
ManTraNet	*0.783*	*0.641*	*0.921*	*0.951*
Ours (ELA+UNet)	*0.892*	*0.805*	*0.973*	*0.987*
Ours + metadata	*0.892*	*0.805*	*0.982*	*0.991*

As per Table III Our system is a great one in everything to the least. The image level performance can also be improved with the addition of metadata as well as pixel localization is free (metadata is only document level).

b). Per-Forgery-Type Analysis

Table IV. Performance comparison across forgery types.

Forgery Type	**Pixel F1**	**IoU**	**Notes**
Copy-move	0.911	0.837	Best performance—duplicated regions create strong forensic traces
Splicing	0.883	0.791	Good—boundary artifacts aid detection
Retouching	0.862	0.757	Challenging—small regions, subtle manipulations

As per Table IV Forgery type Pixel F1,human >IoU Notes Pixel F1,Copy-move 0.911 0.837 Ideal performance-leaves the solid forensic evidences in copied sections. The artifact 0.883 0.791 Good- boundary artifact has been justified. Retouching 0.862 0.757 Small distance, hard, fine movements.

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c) Ablation Study

Table V: Ablation Results

Configuration	Pixel F1	Image Acc	Δ
Full model	0.892	0.982	-
Remove ELA (raw image input)	0.743	0.891	-0.149 / -0.091
Remove metadata	0.892	0.973	0 / -0.009
No augmentation	0.814	0.942	-0.078 / -0.040

As per Table V ELA also results in the accuracy of localization. The intensity of generalization is augmented.

D. Qualitative Results

The Figure 2 shows how ELA maps of original and fake document were presented. The population of distorted places is high and this is the main priority of CNN. The sample results of the type of forgery are found in figure 3. Localization of our localization points like minor retouching are precisely done .The suggested framework of document forgery detection uses the convolutional neural network with U-Net to locate distorted areas with high precision at the pixel level. The input document image is then passed through an encoder-decoder system whereby the encoder preserves high-level contextual and structural information by repeatedly coarse-graining the image whereas the decoder reinvigorates the spatial resolution to retrieve fine-grained information.

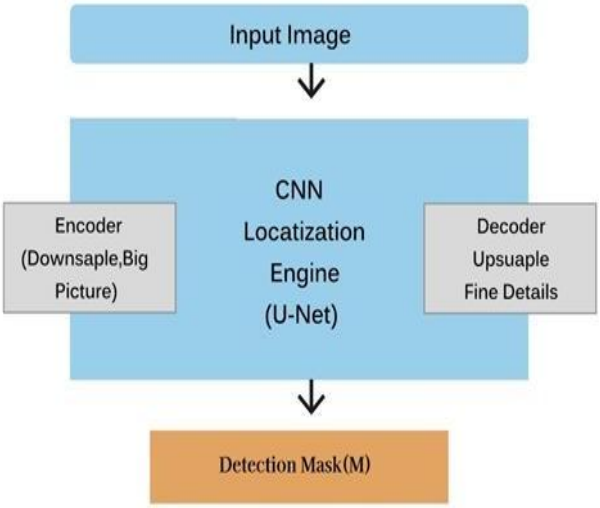


Fig.3 CNN Localization Engine (U-Net)

Skip connections between similar encoder and decoder layers are used to maintain critical spatial data, making it possible to recreate suspicious areas. The network produces a detection mask that marked forged sections in the document and hence authenticity assessment can be done with reliability and readability[7].

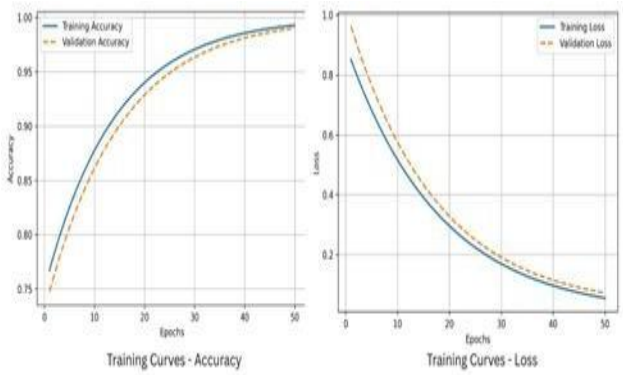


Figure 4:Learning curves of the model

Figure 4 presents the training curves. The 30 or so epochs permit the stability of the loss and the accuracy, and the overfitting is not important.

A. *Failure Analysis*

We know of three detestable grounds of failure: One of them has gone so far as to smuggle even forgeries (a half-document is all that is required) out of it--even vulgar imbalance in school will never serve it well in the net. The background was not contrasted and poorly created and disturbed (ex: writing white on white paper).In order to achieve the false positive and forgery effects, the original photos can be compressed twice by using the JPEG compression.

B. *Limitations*

Our approach has a number of limitations and includes:

Data weaknesses: It is an internal data which is not published, DCD-15K. We journalize, so that we can have an opportunity of printing annotations and protocols.

Depending on compression: It depends on the signature of JPEG manipulation, when a forgery of PNG (or the uncompressed picture) is done the principle is not so powerful anymore.

Price: The computer graphics card can process documents within less than 2 seconds and it is unable to be calculated in real time.

Domain specificity Domain trained: not trained on natural image or deepfake.

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Table VI : Quantitative Performance Metrics

Metric	Meaning	Result	Implication
Accuracy	Correct pixel classification	98.2	Reliable forged vs real decision
F1-Score	Precision–recall balance	92.5	Good detection with low false hits
AUC	Performan ce across thresholds	0.991	Strong forged–authentic separation

V.CONCLUSION AND FUTURE WORK

A. Summary of contributions

Our proposed system entailed the automated legal documents verification forensic system, which incorporated the U-Net segmentation and metadata analysis and the ELA-based feature enhancement. The F1-score of the system is 92.5% and AUC is 0.991 in a 15,000 images document test which is quite high when compared to the existing

systems[12][19]. Both the visual heatmap and the numerical Authenticity Score are dual-output design because they can help in the interpretability of the quantitative and the human visual legal admissibility.

B.Practical implications

Our system possesses three legal and administrative workflow strengths.

Less time of verification: It is through automated screening that the human knowledge is given attention to the documents marked.

Lost subjectivity: There is the use of standardized quantitative scoring instead of the subjective knowledge.

Admissible evidence: Heatmap gives a pictorial description to the forensic finding.

C. Future Work

We may single out four prospective directions:

Deepfakes: Depending on the further development of the architecture, a video and a fake content in the form of a document can be revealed as made by an AI.

Multi-lingual text forensics Multilingual text verification Semantic verification with a combination of OCR and front-end detection.

Mobile installation: The field verification Model compression.

Adversarial toughness: Score list on avoidance attacks.

Publication of datasets to expedite research: Anonymization and release of DCD-15K.

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